

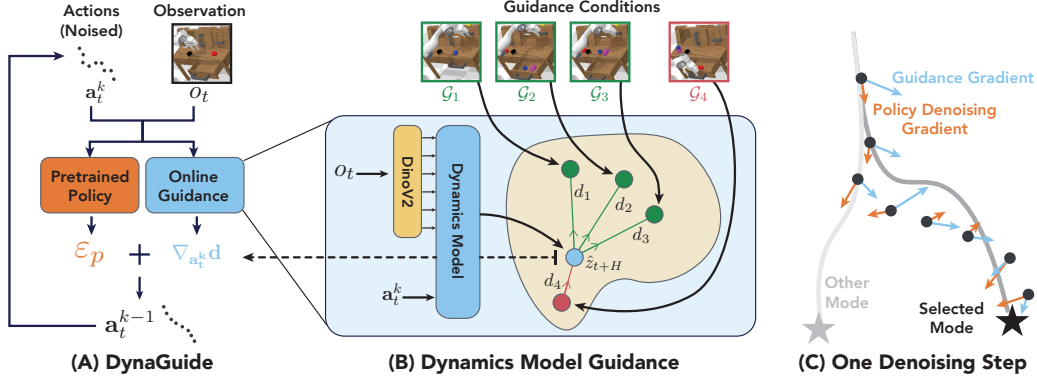


Figure 2: **Achieving Dynamics Guidance.** (A): DynaGuide combines action denoising gradients ϵ_p from the pretrained policy with a guidance gradient $\nabla_{a_t^k} \mathbf{d}$ that increases the likelihood of accomplishing a set of guidance conditions $\mathcal{G}$. (B): Inside the guidance module, a dynamics model predicts future outcomes $\hat{z}_{t+H}$ and compares them to guidance conditions $\mathcal{G}$ (desired / undesired outcomes). We use the latent distances d to define a guidance metric $\mathbf{d}$ (Equation 2) and take the gradient to get the guidance signal $\nabla_{a_t^k} \mathbf{d}$ used by DynaGuide. (C): An example of one denoising step. The pretrained policy seeks behavior modes in the data, while the guidance gradient selects a particular mode.

dynamics model and use the model to approximate these probabilities (§3.1) and then use the gradient of that dynamics model to create π' by modifying its action denoising process (§3.2).

3.1 A Dynamics Model Capable of Guidance

Creating a model to approximate $p(\mathbf{g}^+ | o_t, \mathbf{a})$ and $p(\mathbf{g}^- | o_t, \mathbf{a})$ has two requirements: 1) the model needs to predict future outcome $\hat{o}_{t+H}$ from current observation o_t and action sequence $\mathbf{a}$, 2) we need to compare o_{t+H} to the guidance conditions. Here, H can be a large value or $t+H$ can be the end of the trajectory.

Past works on predicting future observations for planning and reinforcement learning have used *latent dynamics models* to satisfy these two requirements. These dynamics models $h_\theta(\phi(o_t), \mathbf{a})$ operate in a learned latent space $z_t = \phi(o_t)$. The recent work of Dino-WM [57] demonstrated the usefulness of latent distances in the pretrained DinoV2 image embeddings [38] for planning. Taking inspiration from Dino-WM, we leverage the same DinoV2 image latent space for our observation comparisons. Concretely, we use the patch embeddings from DinoV2 as $z_t = \phi(o_t)$ and train a transformer $h_\theta(\mathbf{a}, z_t)$ to predict a latent outcome representation $\hat{z}_{t+H}$. Because ϕ is frozen and we have access to full trajectories of data during training, it is sufficient to train h_θ through a regression objective (Eq. 1). For more details, refer to Appendix B.

$$\mathcal{L}(o_t, o_{t+H}, \mathbf{a}) = \|\phi(o_{t+H}) - h_\theta(\phi(o_t), \mathbf{a})\|_2^2 \quad (1)$$

With this dynamics model trained, we define a *guidance metric* that compares the predicted outcome with the guidance conditions. Given one guidance condition g_i^+ , we want to approximate $p(g_i^+ | o_t, \mathbf{a})$, which can be done by comparing the predicted outcome $\hat{z}_{t+H}$ to g_i^+ . The condition g_i^+ is a visual observation, so we project it into the same latent space $z_i^+ = \phi(g_i^+)$. This allows us to directly compare z_i^+ and $\hat{z}_{t+H}$. If we roughly assume that the latent space is gaussian in structure, a reasonable approximation of $\log p(g_i^+ | o_t, \mathbf{a})$ is proportional to $-||\hat{z}_{t+H} - z_i^+||_2^2$, a metric found in similar forms elsewhere for latent reinforcement learning [55, 34] and planning [57].

Algorithm 1 DynaGuide (Inference-Time)

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1: Input: Guidance Conditions  $\mathbf{g}^+, \mathbf{g}^-$ , Dynamics model  $h_\theta$ 
2: Input: Action denoiser  $\epsilon_\phi(a, o)$ , current obs  $o_t$ 
3:  $\mathbf{a}^K \leftarrow \text{Sample from } \mathcal{N}(0, I)$ 
4: for  $k$  in  $K$  to 1 do ▷ Action Denoising
5:   for  $i$  in 1 to  $M$  do ▷ Stochastic Sampling
6:      $\epsilon \leftarrow \epsilon_\phi(\mathbf{a}^k, o_t)$ 
7:      $\mathbf{d} \leftarrow \text{Eq. 2}$ 
8:      $\hat{\epsilon}(\mathbf{a}^k, o_t) \leftarrow \epsilon - s\sqrt{1 - \bar{\alpha}_k} \nabla_{\mathbf{a}^k} \mathbf{d}$ 
9:     if  $i < M$  then
10:       $\mathbf{a}^k \leftarrow \text{Denoise } \mathbf{a}^k \text{ using } \hat{\epsilon}$ 
11:     else
12:       $\mathbf{a}^{k-1} \leftarrow \text{Denoise } \mathbf{a}^k \text{ using } \hat{\epsilon}$ 
13:     end if
14:   end for
15: end for

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